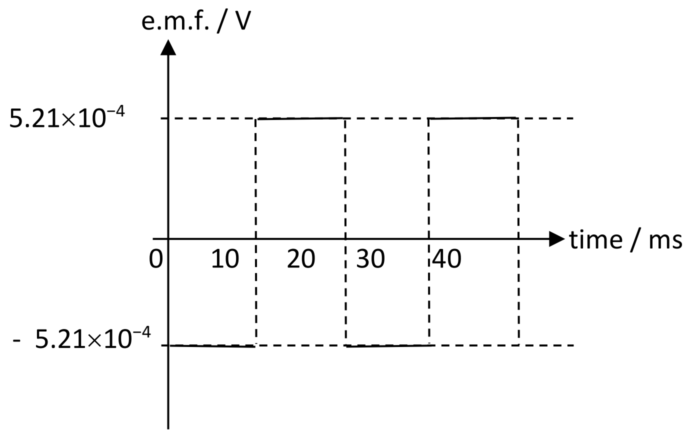
4	**(a)**	Any of 2 of the following: Increase current flowing through the solenoid Increase the number of turns per unit length in the solenoid Insert a ferrous core into the solenoid	 **B1** **B1** **B1** **(Max 2)**
	(b)(i)	C to D	**B1**
	(b)(ii)	By POM, F x 0.106 = 5.7x10^{-4} x 0.077 And F = BIL = B(4.9)(0.025) B = 3.38 x 10^{-3} T Tesla	 **C1** **C1** **A1** **B1**
	(c)(i)	$E = \left\| -\frac{d\Phi}{dt} \right\|$ $= \frac{d(NBA)}{dt}$ $= NA\frac{dB}{dt}$ $= NA\frac{d(\mu_o nI)}{dt}$ $= NA\mu_o n\frac{dI}{dt}$ $= (9)(0.010 \times 0.018)(4\pi \times 10^{-7})(\frac{400}{0.50})(\frac{3.2}{10 \times 10^{-3}})$ $= 5.21 \times 10^{-4}$ V	**C1** **A1**
	(c)(ii)	 Horizontal lie of constant e.m.f. with correct values (allow for e.c.f.) Correct sign of V	 **B1** **B1**

5	**(a)(i)**	$\langle V^2 \rangle = [(3^2 \times 3) + (6^2 \times 1)] / 4$	